Conjecture 8 is proved: Proposition 6.16 (the half-step operator)

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Draft: LG/OG negative-degree paper, 27 pp, WIP commit `b2fb272`
`https://github.com/grandpa-rick/work-in-progress/blob/b2fb2724b2d0dcb14300c62986b79683cc958087/papers/lg-og-negative-degree/main.pdf`

1. What changed

Conjecture 8 of the draft (the half-step operator T) is now Proposition 6.16 (label `prop:T`) in a new §6.5 "The half-step operator" (from p. 21; Prop. 6.16 is on p. 22). The conjecture list in §8 no longer has it. Nothing else in the paper depends on it. The statement is the corrected Wake-257 text. Two things were added: the word "unitriangular" and a uniqueness line.

2. The statement (as printed)

All matrices are indexed by strict partitions with parts at most m and vanish unless $\nu \subseteq \kappa$. By Lemma 6.8(a) (`lem:layers`), $s_{x+1} = A_o A_e\, s_x$. A unitriangular matrix in this algebra has at most one unitriangular square root. Let T be the unitriangular square root of $A_o A_e$.

Proposition 6.16.

(a) $s_{x+1/2} = T\, s_x$ for every $x \in \mathbb{C}$.

(b) Each entry $T(\kappa, \nu)$ is the product of the weights of the connected components of $\theta = D(\kappa) \setminus D(\nu)$. A component D has weight 1 if D is a single off-diagonal box. It has weight 0 in three cases: D has no diagonal box and at least two boxes; D contains at least two diagonal boxes; or D contains exactly one diagonal box and contains a 2×2 square. If D is a ribbon (no 2×2 square) with exactly one diagonal box, its weight is $\varepsilon\,(-\frac{1}{2})^{|D|}$. Here $\varepsilon = +1$ if the left neighbour of the north-east end box of D lies in D, and $\varepsilon = -1$ otherwise. In particular every entry is 0 or $\pm 2^{-k}$.

3. How the proof works

Signed Möbius functions (Lemma 6.17). Let $\zeta(\kappa, \nu) = [\nu \subseteq \kappa]$ and let ζ_o be ζ restricted to $\ell(\nu) = \ell(\kappa)$. Let $S = \mathrm{diag}((-1)^{|\kappa|})$. Then

$$A_e = S\zeta^{-1}S, \qquad A_o = S\zeta_o^{-1}S.$$

The proof is one line: $SA_eS\zeta$ at (κ,ν) is the alternating sum over subsets of the maximal boxes of θ, so it equals $[\theta=\varnothing]$.

Row deletion. For $\nu\neq\kappa$ let r_0 be the first row where they differ. Write $[\nu\prec\kappa]=[\nu_s\le\kappa_{s+1}\ \forall s\ge r_0]$: ν keeps the first r_0-1 parts of κ, deletes the part κ_{r_0}, and fits its tail under the shifted tail of κ. Define

$$X(\kappa,\nu)=\delta_{\kappa\nu}-[\nu\neq\kappa][\ell(\nu)=\ell(\kappa)-1]\,2^{-\kappa_{r_0}}[\nu\prec\kappa],\qquad P(\kappa,\nu)=\delta_{\kappa\nu}+[\nu\neq\kappa]\,2^{-\kappa_{r_0}}[\nu\prec\kappa].$$

So $X=I-$(row deletion at cost $2^{-\text{part}}$, length drop exactly 1). P is the same with a plus sign and no length condition.

Identities (Lemma 6.18). First-row recursions express $X(\kappa,\cdot)$ and $P(\kappa,\cdot)$ through $\kappa'=(\kappa_2,\kappa_3,\dots)$. From these, induction on $\ell(\kappa)$ gives

$$X\zeta=\zeta_oP\quad\text{(c)},\qquad XP=I\quad\text{(d)},$$

so $P=X^{-1}$.

Square root. Let $\tilde T=\zeta_o^{-1}X=P\zeta^{-1}$. Then $\tilde T^2=\zeta_o^{-1}(XP)\zeta^{-1}=\zeta_o^{-1}\zeta^{-1}$. So $(S\tilde TS)^2=A_oA_e$, and $S\tilde TS$ is unitriangular. By uniqueness,

$$T=S\zeta_o^{-1}XS=A_o\,(SXS).$$

(a): Lemma 2.6 gives $s_0(\kappa)=(-1)^{|\kappa|}$, so $Ts_0=S\,P(\cdot,\varnothing)=((-1)^{|\kappa|}2^{-\kappa_1})_\kappa=s_{1/2}$ by Theorem A. T commutes with A_oA_e, which gives $s_{j+1/2}=Ts_j$ for all integers $j\ge0$. Both sides are polynomial in x, so the identity holds for all x. *(b):* Expand $T=A_o(SXS)$ as an inclusion–exclusion over sets of off-diagonal maximal boxes of θ. On length-drop-1 entries, $X(\rho,\nu)=-2^{-(1+c(\rho/\nu))}[\rho/\nu$ connected], where c is the largest content. If the component D containing the diagonal box has a 2×2 square, a sign-reversing toggle at the end of the lowest overlapping row kills the sum. If D is a ribbon, only $\varnothing$ and the north-east end box survive, which gives $\varepsilon(-\frac12)^{|D|}$.

4. Verification

The script `scripts/prove257/verify.py` checks Lemma 6.17, Lemma 6.18(c)(d), $(S\tilde TS)^2=A_oA_e$, the printed rule (b) and $Ts_0=s_{1/2}$. All are true for $m=4,\dots,8$; $m=8$ has 256 strict partitions. For $m\le6$, T also agrees exactly with the binomial series for $(A_oA_e)^{1/2}$. The negative control `ctrl.py` multiplies the X-weight by $3/2$ for $\ell\ge2$, and every identity fails.

5. A cross-check against Theorem C (corrected)

At $x=1$, Lemma 2.6 (`lem:anchor`) gives $s_1=\delta_\varnothing$. So by (a) the $\varnothing$-column $T(\cdot,\varnothing)$ is $s_{3/2}$, and (b) computes it. Here $\theta=D(\kappa)$, whose $\ell(\kappa)$ diagonal boxes all lie in one component. Hence

$$T(\kappa,\varnothing)=\begin{cases}0, & \ell(\kappa)\ge2,\\ (-\frac12)^k, & \kappa=(k),\ k\ge2,\\ \frac12, & \kappa=(1).\end{cases}$$

This is exactly Theorem C (Thm 6.6). So (b) is tested at a non-degenerate index against a theorem with an independent proof. In Theorem C the factor $2^{-\kappa_1}$ is the cost of deleting the row κ_1.

Correction to my 2026-10-25 note. This is *not* the same fact as Theorem A (Thm 6.5), $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$. That one is the column $Ts_0 = SP(\cdot, \varnothing)$ at $x = 0$. Also, $T(\kappa, \varnothing) \neq P(\kappa, \varnothing)$ in general: $T((3), \varnothing) = -\frac{1}{8}$ but $P((3), \varnothing) = \frac{1}{8}$. Commit `b2fb272` adds a remark saying this right after Prop 6.16.

6. Trust grade

Registry `proofs/registry/q488-converse.json`: node `q488-oc-half-step-T-ribbons` is **proved**. Its premises `q488-oc-T-mobius` (Lemma 6.17) and `q488-oc-T-XP-identities` (Lemma 6.18) are both proved and self-contained. No outside citation is used beyond the draft's own Lemma 2.6, Lemma 6.8 and Theorem A. The proof record is `proofs/2026-10-25-conjT-half-step-operator.md`.

7. What I would like from you

1. **Proposition 4.6** (`prop:BRp1`, the $p = 1$ case of BR Cor. 5.6). This line-read is still pending, and it is the outside check I need most.
2. **§6.5**, a read if you have time. It is lower priority, since nothing else in the paper uses it.